

SS-MELSM vs Two-stage HLM

Empirical example: Saeb

Marwin Carmo

```
library(ivd)
library(data.table)
library(ggplot2)
```

The illustration uses openly accessible data from The Basic Education Evaluation System (Saeb) conducted by Brazil's National Institute for Educational Studies and Research (Inep), available at <https://www.gov.br/inep/pt-br/aceso-a-informacao/dados-abertos/microdados/saeb>. It is also available as the `saeb` dataset in the `ivd` package.

Data preparation

Separate within- from between-school effects. That is, besides `student_ses`, compute `school_ses`.

```
## Calculate school-level SES
school_ses <- saeb[, .(school_ses = mean(student_ses, na.rm = TRUE)), by = school_id]

## Join the school_ses back to the original dataset
saeb <- saeb[school_ses, on = "school_id"]

## Define student level SES as deviation from the school SES
saeb$student_ses <- saeb$student_ses - saeb$school_ses

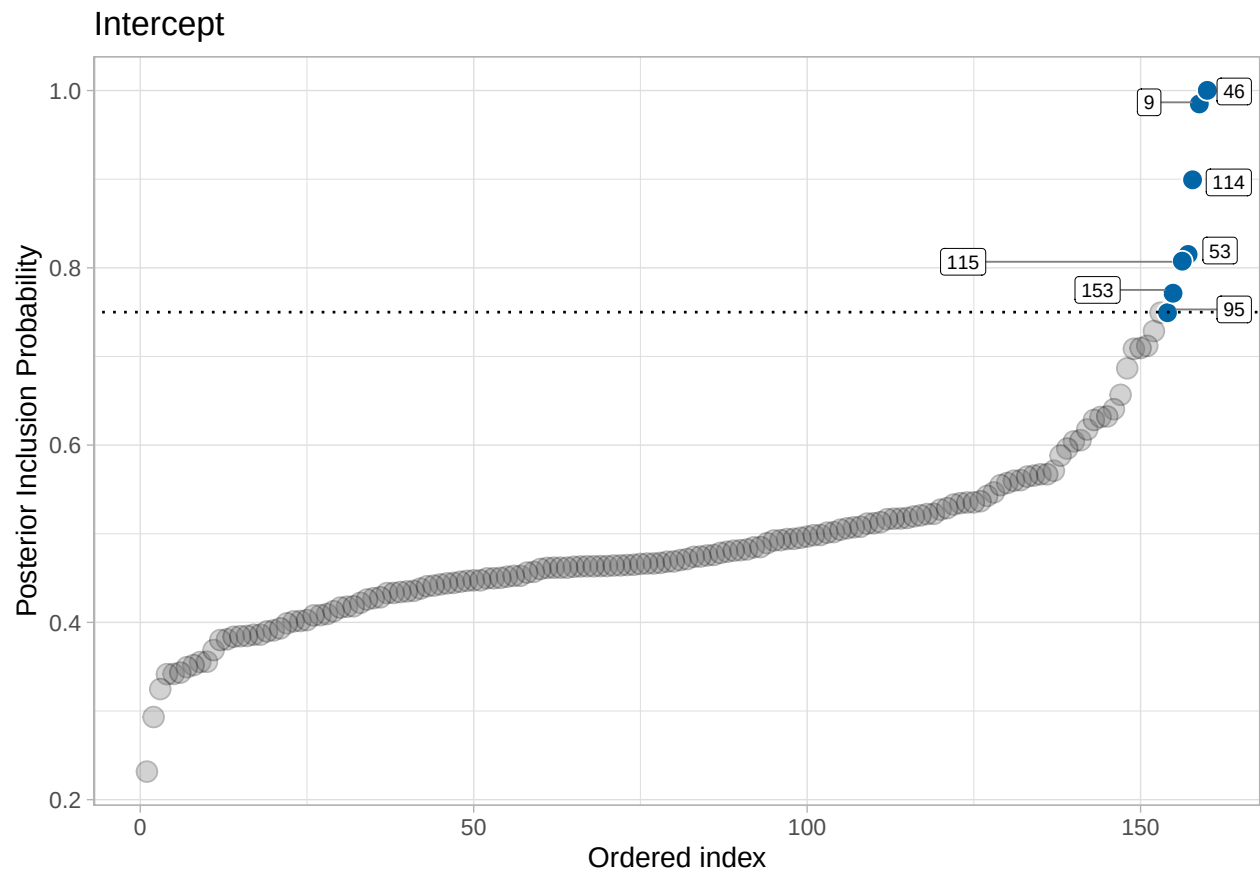
## Grand mean center school ses
saeb$school_ses <- c(scale(saeb$school_ses, scale = FALSE))
```

Fit SS-MELSM with ivd

```
out <- ivd(location_formula = math_proficiency ~ student_ses * school_ses + (1|school_id),
  scale_formula = ~ student_ses * school_ses + (1|school_id),
  data = saeb,
  niter = 3000, nburnin = 12000, WAIC = TRUE, workers = 6)
```

Posterior Inclusion Probabilities

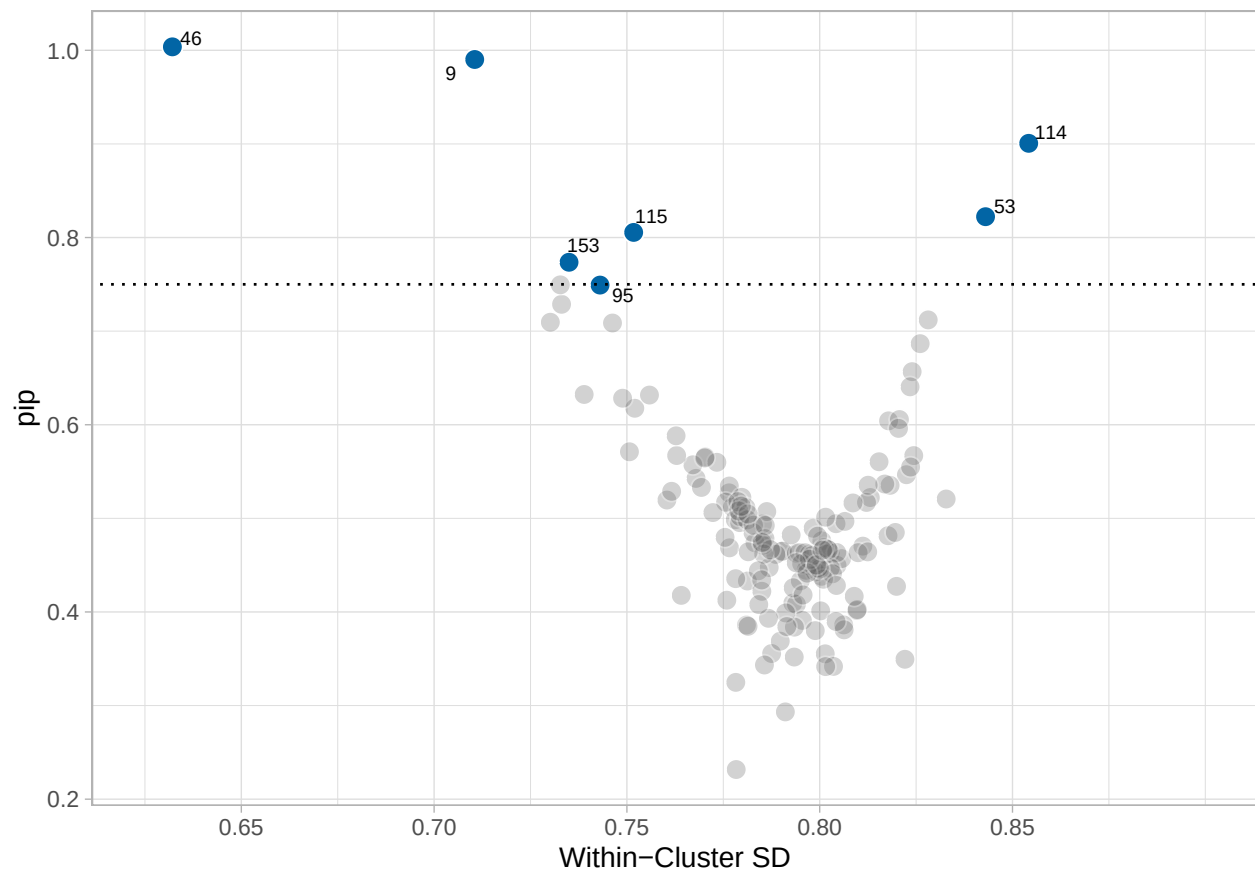
```
plot(out, type = "pip") +
  theme_light(16)
```



Within-cluster SD

```
p2 <- plot(out, type = "funnel")
```

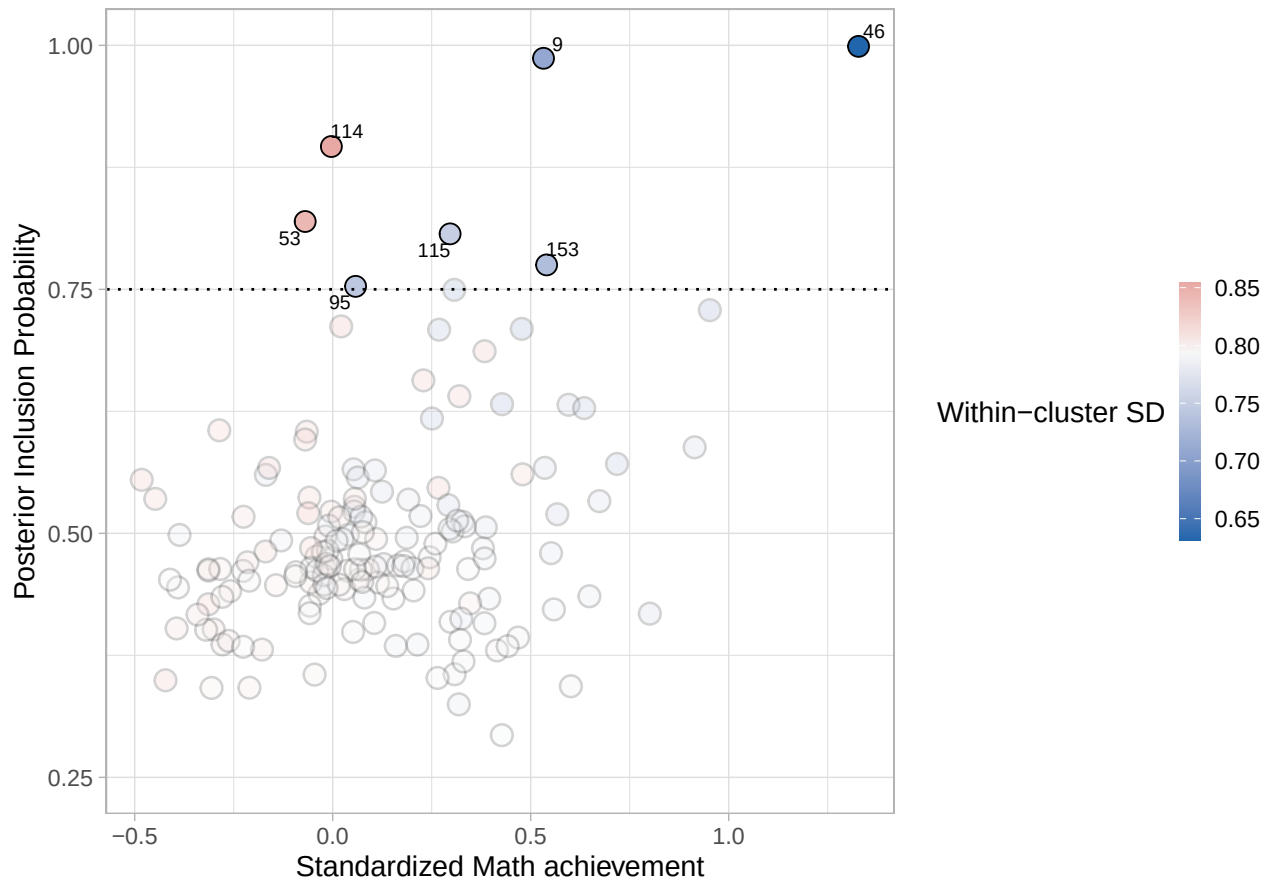
```
p2 +  
  theme_light(16) +  
  scale_x_continuous(limits = c(0.625, .90),  
                     breaks = seq(0.65, .85, by = 0.05))
```



Math achivement

```
p3 <- plot(out, type = "outcome")
```

```
p3 +  
  theme_light(16) +  
  labs(  
    x = "Standardized Math achivement") +  
  scale_y_continuous(limits = c(0.25, 1.0), breaks = seq(0.25, 1, by = 0.25))
```



Caterpillar plot

```
# We'll need to use `.summary_table()` and `.extract_to_mcmc()`, which are internal
# functions from `ivd`, that are not meant for direct use

col_names <- dimnames(ivd:::.summary_table(out$samples[[1]]$samples ))[[2]]
Kr <- out$nimble_constants$Kr

## find scale random effects
## Extract numbers and find locations
column_indices <- sapply(col_names, function(x) {
  if (grepl("^u\\[", x)) { # Check if the name starts with 'u['
    ## Extracting numbers
    nums <- as.numeric(unlist(strsplit(gsub("[^0-9,]", "", x), ",")))
    ## Check if second number (column index) is greater than Kr
    return(nums[2] > Kr)
  } else {
    return(FALSE)
  }
})

scale_ranef_pos <- which(column_indices)
posterior_samples <- do.call(rbind, lapply(ivd:::.extract_to_mcmc(out),
  FUN = function(x) x[, scale_ranef_pos]))
```

```

## Find the largest residual SD
## Extract the posterior mean of the fixed effect:
zeta <- mean( unlist( lapply(ivd:::extract_to_mcmc( out ), FUN = function(x) mean(x[, "zeta[1]"]))) ) )
## Extract the posterior mean of each random effect:
u <- colMeans(posterior_samples)

tau <- exp(zeta + u )

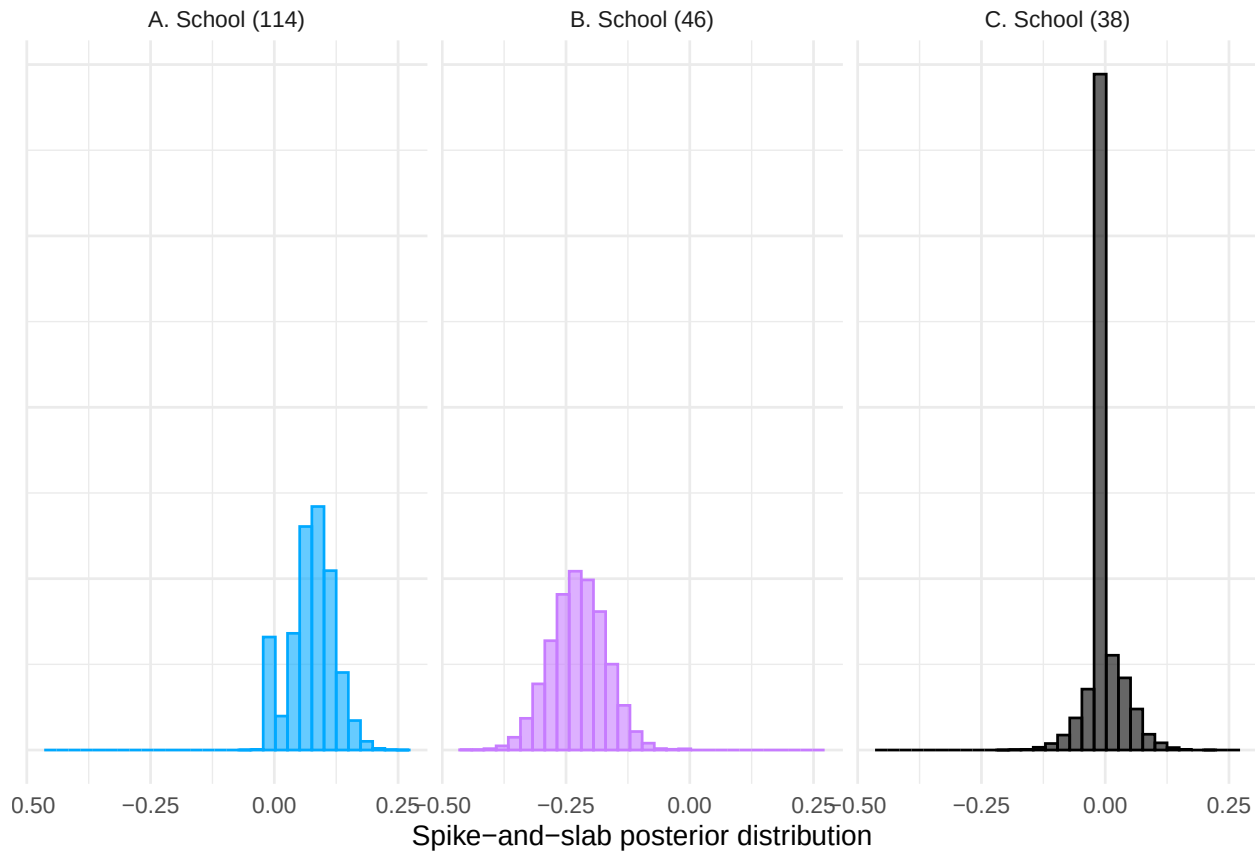
## find the median
mid_indices <- c(160/2, 160/2 + 1)
sorted_vec <- sort(tau)
middle_values <- sorted_vec[mid_indices]

df_hist <- data.frame(
  min = posterior_samples[, which(tau == min(tau))],
  median = posterior_samples[, which(tau %in% middle_values[1])],
  max = posterior_samples[, which(tau == max(tau))]
)

facet_labels <- c("max" = "A. School (114)", # with the largest residual SD",
  "min" = "B. School (46)", #, with the smallest residual SD",
  "median" = "C. School (38)" # with median residual SD"
)

df_hist |>
  tidyr::pivot_longer(cols = dplyr::everything(),
    names_to = "text",
    values_to = "posterior") |>
  dplyr::mutate(text = factor(text, levels = c("max", "min", "median"))) |>
  #dplyr::mutate(text = forcats::fct_reorder(text, posterior)) |>
  ggplot(aes(x=posterior, color=text, fill=text)) +
  geom_histogram(alpha=0.6) +
  scale_fill_manual(values=c("#00A9FF", #114
    "#C77CFF", #46
    "#000000" #38
  )) +
  scale_color_manual(values=c("#00A9FF", #114
    "#C77CFF", #46
    "#000000" #38
  )) +
  facet_wrap(~text,
    labeller = labeller(text = facet_labels)) +
  theme_minimal(16) +
  guides(color = "none", fill = "none") +
  theme(
    axis.text.y = element_blank(),
    axis.title.y = element_blank()
  ) +
  labs(x = "Spike-and-slab posterior distribution")

```



Caterpillar plot

```
col_names <- dimnames(ivd:::.summary_table(out$samples[[1]]$samples ))[[2]]
Kr <- out$nimble_constants$Kr
cols_to_keep <- c( )
## Find spike and slab variables
col_ss <- col_names[ grepl( "^ss\\[", col_names ) ]
for(col_name in col_ss) {
  col_number <- as.numeric(unlist(regmatches(col_name, gregexpr("[0-9]+", col_name)))[1])
  if(col_number > Kr ) {
    cols_to_keep <- c(cols_to_keep, col_name )
  }
}
## Subset each MCMC matrix to keep only the relevant columns
subsamples <- lapply(ivd:::.extract_to_mcmc(out), function(x) x[, cols_to_keep])
## Calculate column means for each subsetted MCMC matrix
means_list <- lapply(subsamples, colMeans)
## Average across the lists and chains
final_means <- Reduce("+", means_list) / length(means_list)
## assign the means to the specific random effects
ss_means <- list()
## Select the ss effects
Sr <- out$nimble_constants$Sr
for(i in 1:Sr ) {
```

```

index <- paste0("\\\\", i+Kr)
position_ss_value <- grepl(index, names(means_list[[1]]))
ss_means[[i]] <- final_means[position_ss_value]
}

df_pip <- data.frame(id = seq_len(length(ss_means[[1]])),
                    pip = ss_means[[1]])

u_summary <- data.frame(
  RandomEffect = 1:length(u),
  Median = apply(posterior_samples, 2, function(x) quantile(x, probs = 0.5)),
  Lower95CI = apply(posterior_samples, 2, function(x) quantile(x, probs = 0.025)),
  Upper95CI = apply(posterior_samples, 2, function(x) quantile(x, probs = 0.975)),
  pip = df_pip$pip
)

u_ordered <- u_summary[order(u_summary[, "Median"]), ]
u_ordered$ordered <- 1:nrow(u_summary)

ggplot(u_ordered, aes(x = ordered, y = Median)) +
  geom_pointrange(
    aes(ymin = Lower95CI, ymax = Upper95CI),
    size = 0.7,
    linetype = "solid",
    color = "black",
    alpha = 0.7
  ) +
  geom_point(data = subset(u_ordered, pip < 0.75),
            size = 2.5, stroke = .1, aes(size=tau), alpha = .3,
            shape = 21, fill = "black", color = "black") +
  geom_point(data = subset(u_ordered, pip >= 0.75),
            aes(fill = factor(RandomEffect)), size = 2.5, stroke = .1,
            shape = 21, color = "black") +
  geom_abline(intercept = 0, slope = 0, lty = 3) +
  labs(color = "School ID", fill = "School ID", y = "Median Random effect estimate") +
  theme_minimal(16)

```

